Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name: Termini

Team leader name: [to be filled]

Problem Statement: Future of Work & Enterprise Productivity — every organisation should be able to run AI on data it cannot send to a cloud.

Brief about the idea

One clinic, one rule

A dental clinic in Korea wanted AI for its phones, its insurance claims and its own reporting. Sending patient data to a cloud was not an option — so the owner, a biology graduate rather than a computer scientist, built the infrastructure himself.

“All client data must not leave the building.”

The rule the whole build was organised around

Phones, insurance claims, reporting and document retrieval — all running on models in the same room as the chair.

6 months

of evenings, on top of dentistry — the cost of there being no product

0

client records sent off-site; phones, claims and reporting all run in the back room

Source: arca.live/b/aihomelab/185810092 — “A neighbourhood dental clinic's AX transformation”, 13 Oct 2026

Brief about the idea

What can't leave the building

The clinic's rule is not a dental rule — it is the rule of every business whose best data is also its most restricted.

Healthcare & dental
Patient records, calls, insurance claims
Front desk, adjudication, notes

Banking & insurance
KYC files, transactions, loss records
Fraud triage, credit memos, claims

Legal
Privileged matter files, client advice
Contract review, discovery, drafting

Manufacturing
Design files, process parameters, yields
Quality, maintenance, planning

Life sciences
Trial data, patient histories
Protocol analysis, safety review

Public sector & defence
Classified material, citizen records
Analysis, briefings, casework

Software
Source code, incident data, architecture
Code agents, on-call triage

Retail & commerce
Customer PII, supplier terms, margins
Demand, service, pricing

Eight industries, one constraint: the data that would make AI useful is the data that can't be handed over.

And the result: **95%** of GenAI pilots show no measurable P&L impact (MIT NANDA 2025, contested) while **88%** of firms use AI somewhere (McKinsey 2025) — permission, operations and context are missing at once.

Sources: MIT NANDA, The GenAI Divide (Jul 2025) · McKinsey, The State of AI 2025

Brief about the idea

Data used without permission

The loud concern is AI replacing existing labour. The governing one arrives first: the work leaves, and comes back as someone else's product.

Apr 2023 · Samsung

Engineers debugged with ChatGPT

Confidential source code, equipment yield data and a meeting's notes went in as prompts. Samsung restricted generative AI across its network weeks later.

“Data is transmitted and stored to an external server, making it impossible for the company to retrieve it.” — Samsung memo

fortune.com · 2 May 2023

Jan 2025 · DeepSeek

A database of everyone's prompts

Wiz Research found an unauthenticated ClickHouse instance holding 1M+ log lines, chat history, API keys and plaintext passwords.

Not a breach of the model — a copy of every conversation, reachable from the open internet.

wiz.io/blog · 29 Jan 2025

Sep 2026 · Navier–Stokes

The credit left with the data

OpenAI announced a Navier–Stokes solution its team described as using “very little human input”. Twelve hours earlier, Buckmaster said his and Alpöge's unpublished advances had leaked to the lab.

A Millennium Prize result — and a priority dispute about whose work the model had read.

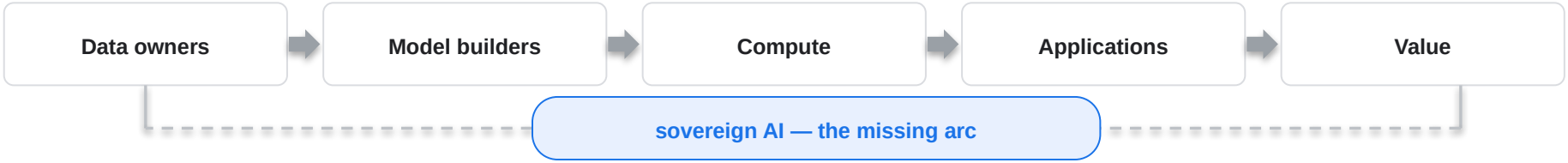
wikipedia.org · Sept 2026

How Termini closes it: not a better policy, but a topology where the copy never leaves — C2 stays on the node, and what does leave is named, classified and audited.

Brief about the idea

The loop only closes where the data lives

A system improves by learning from its own outcomes. When the record lives in someone else's cloud, the loop belongs to someone else.



A memory that stays

Outcomes, corrections and exceptions remain with the entity.

A jury, not an oracle

Several models read the same evidence; disagreement is where a person looks.

A corpus no one else has

The private record no model has seen and no vendor could be given.

Intelligence should be plural. **Governance is not the tax on the loop — it is what closes it.**

Opportunities

Right-size the intelligence

Frontier models are not the answer to every task. They are the top of a ladder — and the product is the routing between the levels.

Level	Runs on	Good at	Data movement	In the demo
L0 · deterministic	Controller rules	Known remedies, threshold-to-action mappings	None	Yes
L1 · small local	The node — 4B class, llama.cpp	Classify, extract, summarise, triage	None	Yes
L2 · large local	A GPU node over your data	Reasoning specific to your domain data	None	Config only
L3 · frontier	Gemini, via Vertex AI	The hardest reasoning, the smallest payload	Minimised C0, if permitted	Yes

L1 triages → L3 reasons, only where policy allows → L0 executes → **the audit records which level answered.**

Local is not worse — it is the right level for most of the work.

Opportunities

Delegate the operations, not the data

That split is the whole proposition. Packaging, partners and pricing all follow from it.

What is delegated

Bootstrap, remediation, upgrades, the on-call rotation. This is the work a small firm cannot staff and a hardware owner does not want to build.

What is never delegated

C2 payloads, transcripts, records, credentials. The data stays on the node; the controller sees only what the classification allows.

Who buys which half

The SME buys the outcome. The hardware owner licenses the layer so its metal is sellable as AI. Both get the same control plane.

One control plane, two buyers, one boundary — **and the boundary is what makes delegation signable.**

List of features offered by the solution

Bootstrap. Operate. Prove.

The clinic needed all three, so one man built all three by hand. In Termini they are one product.

01 · Bootstrap

Hardware becomes a fleet

A bare node boots the pre-baked image, registers in 150 ms, is assigned a role and joins the cluster. No installs, no specialist on site.

Removes: the wiring weekend

02 · Operate

An operator you don't hire

An AI agent watches telemetry and makes decisions the way a data-centre operator would — restarting a runaway workload, stopping what will fill a disk — and falls back to a fixed rule when unsure.

Removes: the on-call rotation

03 · Prove

A boundary you can audit

C0 / C1 / C2 classification, an on-node redactor and a fail-closed controller gate. What leaves is named, classified and logged.

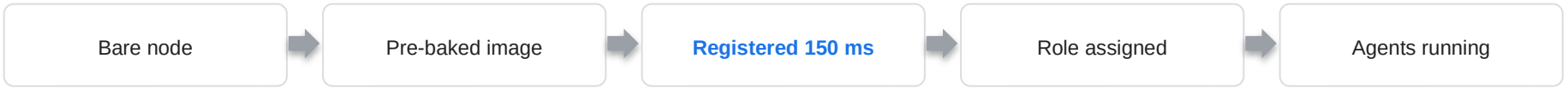
Removes: the compliance gap

The clinic is the requirement. **Termini is the product.**

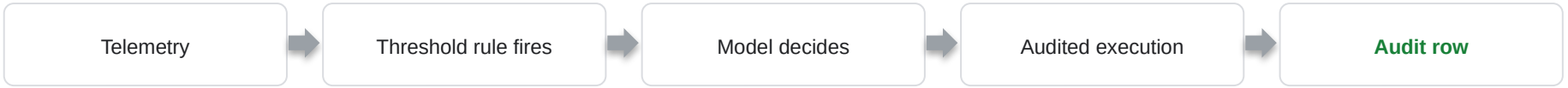
Process flow diagram or Use-case diagram

From bare node to a governed fleet

DAY-0 · BRING-UP



DAY-2 · SELF-HEALING LOOP



The rule detects; the model decides; the executor acts; the audit row links **trigger → decision → action → result.**

Nothing is decided by a model that a person could not read back afterwards — and the deterministic path needs no model at all.

The action space is grammar-constrained: an off-allowlist action has no valid encoding, so safety is enforced in the decoder rather than checked afterwards.

Architecture diagram of the proposed solution

The architecture is the governance claim



Substrate and channel — adopted, not invented

Ubuntu and an LXD cluster across nodes; one outbound-only mTLS WebSocket per node carries heartbeat, telemetry, commands and acks. No route from an agent VM to the controller.

Boundary and loop — the two things we build

A fail-closed telemetry gate that rejects a non-conforming batch rather than dropping fields, and a governed operator loop that audits every action it takes.

EU AI Act Art. 10 / 12 / 14 / 15 · GDPR Art. 25 / 32 / 44–49 · ISO/IEC 42001 · 27001 · SOC 2 Type II · HIPAA §164.312

We don't certify you — we generate the evidence a certification asks for, continuously.

Architecture diagram of the proposed solution

What leaves the building, exactly

Class	Leaves?	Contents
C0	Yes — allowlisted	Hardware and OS health: CPU, RAM, disk, GPU, inference liveness, instance inventory, rule metrics
C1	Yes — transformed	Salted hashes, bounded display labels, timestamps. Never raw identifiers
C2	Never	Workload payloads, call transcripts, patient or customer records, prompts, credentials, logs

The one external call — disclosed

The operator's diagnosis runs on Gemini via Vertex AI. The prompt is C0 only, the audit row records egress=external, and one variable (backend=local) puts it back on the node's own weights.

Why publish the exception

A boundary you cannot see is not a boundary. The claim is not “nothing leaves” — it is “nothing C2 leaves, and everything that does is named and audited.”

Technologies to be used in the solution

Gemini decides. Gemma stays home.

Gemini 2.5 Flash · Vertex AI

The diagnosis — naming the remediation from C0 telemetry. Measured 1.8 s warm. The only call that leaves the premises, and it says so in the audit trail.

Gemma 3 · on the node

Google's open weights, served by llama.cpp. Automatically detects the inference node, bootstraps the toolchain and leaves it ready for agents — nothing forwarded.

Cloud Run · GPU reference

The same Gemma on 1× L4 with vLLM — deployed as the managed reference and the shape this takes when GPU quota stops being the blocker.

GCE controller · GCE node · Firebase Hosting · Cloud Storage (weights) · pre-baked GCE image · LXD · llama.cpp

Local data uses local models. **Where the customer permits it, Termini integrates with Gemini.**

Prototype Performance report/Benchmarking

What runs today, measured

150 ms
Node registration from the pre-baked image — zero installs at boot

0
C2 payloads the controller has ever stored — fail-closed by design

422
A C2 field rejected before any byte is stored, checked three ways

1
Remediation decided by the model, executed and audited end to end

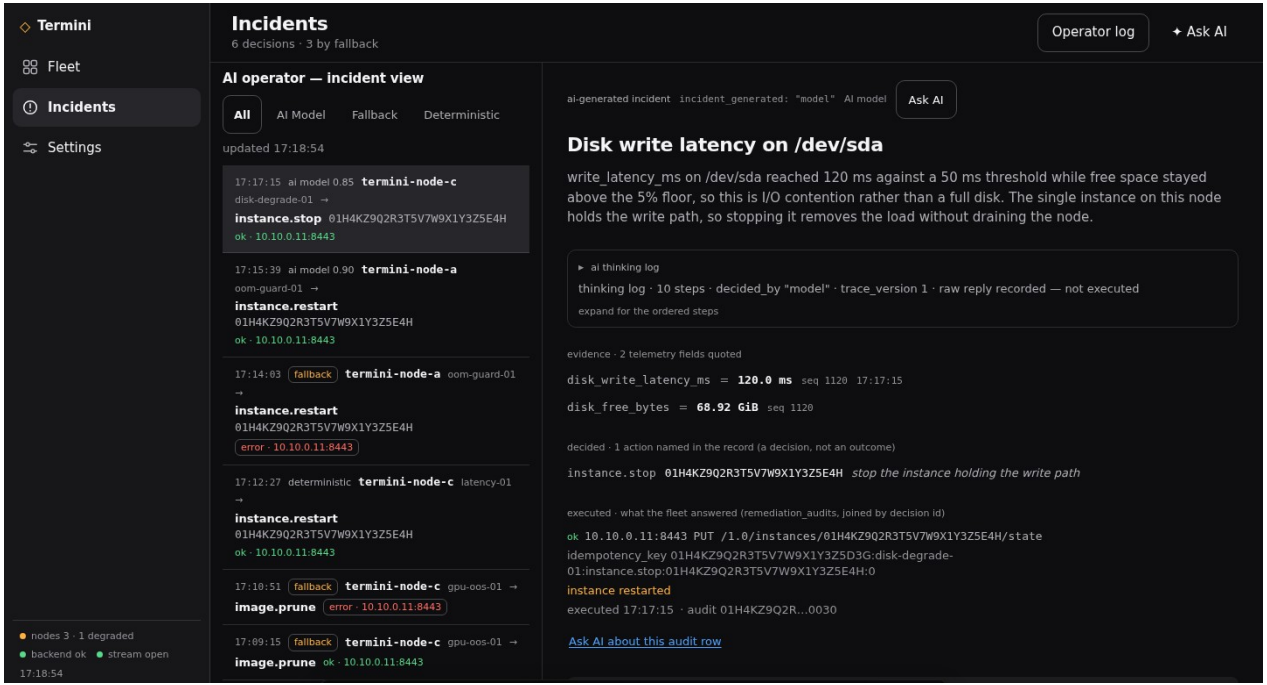
1.8 s
Diagnosis warm on the managed path — 402 s on local CPU weights, stated not hidden

live
On GCE + Firebase + Vertex AI — a judge can open it and watch

Three claims, six measurements. **Nothing on this page is a projection.**

Snapshots of the prototype

The operator, on screen



Operator log

Every incident: the rule that fired, the model's reasoning, the evidence fields, the action executed and the audit id.

Fallback visible

Decisions the model did not take are labelled deterministic or fallback, in the same list.

Live

termini-510609.web.app — the demo is paused between judging sessions; the resume command is in SUBMISSION.md §3.

Additional Details/Future Development (if any)

From one clinic to a fleet of them

Real bare-metal onboarding

MAAS + iPXE netboot, so a box arrives from a partner and joins without a technician on site.

A GPU node in the demo

L4 quota is the current blocker; the path is already built and served on Cloud Run.

Learned baselines

Edge anomaly detection replacing static thresholds — the operator learns each site's normal.

Federated eval exchange

Evals and weights travel between entities; records don't. The plural-ecosystem mechanism, productised.

Compliance mapping

GDPR, EU AI Act, Korea PIPA, India DPDP posture mapped onto the C0/C1/C2 boundary.

Hardware partner channel

The box is commodity; the operator is the product. Partners sell metal, we run it.

Provide links to your:

Links

GitHub public repository	github.com/yanksyoon/termini
Demo video (3 minutes)	script ready — recording is the pending human step (SUBMISSION.md §5)
Final product link	termini-510609.web.app
Live API	34-93-190-64.sslip.io/v1

The dashboard opens with no credential in the URL; sign in with the project token once (Settings → sign in).

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AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

The doctor built it alone.
The next one won't have to.

Your hardware. Your data. Our operations.

